Unit Plan: Nuclear Chemistry

FOR THE TEACHER

Summary
The AACT high school classroom resource library and multimedia collection has everything you need to put together a unit plan for your classroom: lessons, activities, labs, projects, videos, simulations, and animations. We constructed a unit plan using AACT resources that is designed to teach nuclear chemistry to your students.

Grade Level
High School

NGSS Alignment
The teaching resources used in this unit plan will help prepare your students to meet the performance expectations in the following standards:

- **HS-PS1-8:** Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay.
- **HS-PS3-2**: Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects).
- **HS-ESS1-5:** Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks.
- **HS-ESS2-1:** Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features.
- **HS-ETS1-1:** Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.
- **Scientific and Engineering Practices**:
    - Developing and Using Models
    - Analyzing and Interpreting Data
    - Obtaining, Evaluating, and Communicating Information
    - Using Mathematics and Computational Thinking
    - Engaging in Argument from Evidence

Objectives
By the end of this unit, students should be able to

- Compare and contrast the processes of fission and fusion.
- Define isotopes and explain the differences and similarities in subatomic particles for isotopes of the same element.
- Describe what characteristics of an isotope tend to make it radioactive.
- Explain the stabilizing function of neutrons in the nucleus.
- Recognize the factors that contribute to isotopes.
- Calculate the number of subatomic particles in different isotopes.
- Calculate an element's atomic mass based on relative abundance and mass numbers.
- Define stable isotope.
- Explain the difference between a chemical reaction and a nuclear reaction.
- Describe the changes that occur in an atom in three types of nuclear decay.
- Define the meaning of "half-life."
- Indicate what happens to a sample size when it undergoes a half-life decay.
- Identify that radiation is emitted during the decay process.
- Determine the approximate sample size that will remain based on the length of a half-life for a given sample.
- Identify the shape of the graph produced during exponential decay.

- Determine the half-life of a radioisotope from a mass versus time graph.
- Understand how scientists use half-lives and carbon dating to try to determine the age of fossils and rocks.
- Model the process of radioactive decay to illustrate half-life and conservation of mass.
- Model the process using half-life through manipulatives.
- Consistently use radiometric data to date rock samples and explain how radiometric dating works.
- Use radiometric data to suggest a theory about the Atlantic Ocean's past (specifically, seafloor spreading).
- Explain how different radioactive isotopes (with different half-lives) can be used to explore the history of different objects. Specifically, students will consider the difference between C-14 dating of organic material and K-40 dating of seafloor rocks.
- List uses for radioactive isotopes.
- Devise a technique that will collect data, by means of working backwards, to determine the time (time-zero) that the ice began melting at their lab stations.
- Determine the rate of melting.
- Construct a linear graph to illustrate melting rate.
- Construct an exponential graph, utilizing class data, to illustrate melting rate.

Chemistry Topics

This unit supports students' understanding of

- Nuclear Fusion
- Nuclear Fission
- Nuclear Equations
- Origin of Elements
- Radioactive Decay
- Radioactive Isotopes
- Half-Life
- Radiation

Time

Teacher Preparation: See individual resources.
Lesson: 4-6 class periods, depending on class level.

Materials

- Refer to the materials list given with each individual activity.

Safety

- Refer to the safety instructions given with each individual activity.

Teacher Notes

- Some of the resources in this unit plan were written for AP Chemistry. However, most can easily be adapted for an on-level or honors class.
- The activities shown below are listed in order that they should be completed.
- The teacher notes, student handouts, and additional materials can be accessed on the page for each individual activity.
- Please note that most of these resources are AACT member benefits.

Classroom Resources

- Start the unit with the activity, Physical, Chemical, and Nuclear Changes. It will help students understand how different the energy involved in a nuclear reaction is when compared to physical and chemical changes.

- Next, students should read the article *"Where do Chemical Elements Come From?"* from the October 2009 issue of *ChemMatters* magazine. This article discusses the process of nuclear fusion in stars and during supernovae to produce more massive elements. It is a great foundational start to the unit because understanding where all nuclei come from is essential to the discipline of chemistry. As a related challenge, consider tasking students with writing nuclear equations for select fusion reactions that occur on the sun. The October 2009 *ChemMatters* Teacher Guide contains questions to go with the article.
    - It is suggested that teachers read through the Teacher Background section in the October 2009 *ChemMatters* Teacher Guide. This will provide more insight on the life cycle of a star and its relation to chemical elements, as well as other ideas you can use in the classroom to explain nuclear fusion in stars.

    - As an extension, you can have your students read the article *What 'Uuo'ught to Know About Elements 112-118*. It talks about the formation of manmade elements.

- The activity, The Universe of Elements fits in well with discussing fusion in stars. It provides students with a model to conceptualize the abundance of elements in the universe.

- Follow this with the activity, Fission vs. Fusion reading. After reading about the process of nuclear fusion in stars, this will clarify what nuclear fusion is and introduce nuclear fission. The activity includes a graphic organizer with a "text-in-the-middle" reading strategy for students to interact with the text.

- Radioactive decay is a key component of understanding the chemistry of the nucleus. The activity, Why are Some Isotopes Radioactive? has students use a PhET simulation to explore stable versus unstable nuclei and then explains how the band of stability can be used to determine whether nuclei are stable or not. Pair this with the activity, Nuclide Stability Investigation, which allows students to explore the stability of different nuclides and the types of decay they will undergo.

- Follow up with the video-based activity, "What are Isotopes?" produced by the American Chemical Society. The video also explains radioactive decay and talks about the main forms of decay. The link has an accompanying worksheet for students to answer questions during the video.

- The Isotope Sisters Puzzle is a fun way to challenge students after their introduction to isotopes. Students pair isotopes with descriptions of some of their properties to complete a crossword puzzle.

- The activity, Building a Nuclide also fits in well here. Students build nuclides using magnetic balls and create their own definitions for the strong nuclear force and nuclear binding energy using observations they make.

- Now, introduce specific types of decay using the Nuclear Decay Investigation. Students make observations from a decay chain, focusing on how the mass and atomic number change during radioactive decay.

- Use the Radioactive Decay Simulation and supplemental activity to cement knowledge of radioactive decay and allow students to practice identifying the type of decay and even representing decay using chemical equations.

- If you have time to fit in one more activity on decay, use the activity, Nuclear Decay of Uranium. The combination of modeling, writing decay equations, and interpreting graphs makes this activity a great steppingstone to many of the skills needed to master nuclear chemistry.

- There are many activities that can be used to introduce half-life to your students. They all involve similar manipulatives to model the decay of radioactive samples. You can choose from the Using Dice to Explore Radioactive Decay, the Twizzler Half-Life, the Investigating Exponential Decay (with skittles), or the Half-Life activity, using pennies. At the end of this activity, students will have generated an exponential graph that shows what decay looks like.

- Help your students to further visualize half-life with the Half-Life Investigation Simulation and accompanying activity which shows the decay of two different radioactive samples over time.

- The Nuclear Medicine Half-lives activity is a standout in the AACT library. It encourages critical thinking about how and why nuclear medicine is transported and stored the way it is. It also helps students to visualize what happens when nearly all of a substance has decayed away.

- The ACS Landmark lesson plan, Radiocarbon Dating and Willard Libby contains applications of half-life and nuclear reactions. Students will apply their knowledge of radioactive decay to carbon dating and learn about interpreting and writing nuclear equations. The background on carbon dating will prepare students for the radioactive decay lab.

- Depending on time, you may want to have your students complete both analysis activities listed here:
    - The lab, Radioactive Dating: The Demise of Frosty has students collect data on how fast an ice cube is melting and extrapolate a graph to find out when the ice cube was completely frozen. The graph generated from all the student data will model an exponential decay curve as a comparison to the process of radiocarbon dating.

    - The activity, Radioactive Decay and Peat Bogs, compares the amount of C-14 remaining in a peat sample to determine the age of the layer of plant material. Note that the author of this activity does recommend first completing lab, Radioactive Dating: The Demise of Frosty, to introduce the concept of exponential decay in a hands-on way.

Culminating Activities

- One way to culminate the unit is to involve students in a project or debate about nuclear energy. This will allow them to apply their knowledge of nuclear fission and chain reactions and understand that having a scientific understanding can help them cast a more educated vote. The Nuclear Energy debate uses class time to have students defend their point of view to classmates with the opposite point of view, while the Nuclear Energy Power Plants classroom resource has them write a persuasive essay expressing their point of view.

- The activity, Radiological Applications of Isotopes has students research a chosen radioactive isotope and discuss its potential to treat medical problems. Students also apply their knowledge of radioactive decay by filling in properties of either the parent or daughter isotope.

- Using Stable Isotopes to Determine Material Origin is an activity that offers a great way to review some of the major concepts learned in the unit. Students will review isotopes as they compare atoms of the same element by identifying the number of protons and neutrons they have and then use that information to complete some isotopic analysis.